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Asset management — Overview, principles and terminology

Gestion d'actifs — Aperçu général, principes et terminologie

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the WTO principles in the Technical Barriers to Trade (TBT) see the following URL: Foreword - Supplementary information

The committee responsible for this document is Project Committee ISO/PC 251, *Asset management*.

This corrected version of ISO 55000:2014 incorporates the correction of the numbering of definitions 3.2.4 to 3.2.7 and corresponding cross-references.

Introduction

0.1 Purpose

This International Standard provides an overview of asset management and asset management systems (i.e. management systems for the management of assets). It also provides the context for ISO 55001 and ISO 55002.

International cooperation in the preparation of these standards has identified common practices that can be applied to the broadest range of assets, in the broadest range of organizations, across the broadest range of cultures.

0.2 Relationship with other standards

ISO 55001, ISO 55002 and this International Standard relate to a management system for asset management, which is referred to as an “asset management system” throughout the three standards.

ISO 55001, ISO 55002 and this International Standard can be used in combination with any relevant sector or asset type-specific asset management standards and technical specifications. ISO 55001 specifies requirements for an asset management system, while the other standards detail sector-specific, asset-specific or activity-specific technical requirements or give guidance on how ISO 55001 should be interpreted and applied within a specific sector or to particular asset types.

0.3 Target audience

This International Standard is primarily intended for use by:

- those considering how to improve the realization of value for their organization from their asset base
- those involved in the establishment, implementation, maintenance and improvement of an asset management system
- those involved in the planning, design, implementation and review of asset management activities; along with service providers.

0.4 Benefits of the standards

The adoption of ISO 55001, ISO 55002 and this International Standard enables an organization to achieve its objectives through the effective and efficient management of its assets. The application of an asset management system provides assurance that those objectives can be achieved consistently and sustainably over time.

[Annex A](#) provides additional information on areas related to asset management activities.

[Annex B](#) shows the relationship between key elements of an asset management system.

Asset management — Overview, principles and terminology

1 Scope

This International Standard provides an overview of asset management, its principles and terminology, and the expected benefits from adopting asset management.

This International Standard can be applied to all types of assets and by all types and sizes of organizations.

NOTE 1 This International Standard is intended to be used for managing physical assets in particular, but it can also be applied to other asset types.

NOTE 2 This International Standard does not provide financial, accounting or technical guidance for managing specific asset types.

NOTE 3 For the purposes of ISO 55001, ISO 55002 and this International Standard, the term “asset management system” is used to refer to a management system for asset management.

2 Asset management

2.1 General

The factors which influence the type of assets that an organization requires to achieve its objectives, and how the assets are managed, include the following:

- the nature and purpose of the organization;
- its operating context;
- its financial constraints and regulatory requirements;
- the needs and expectations of the organization and its stakeholders.

These influencing factors need to be considered when establishing, implementing, maintaining and continually improving asset management.

Effective control and governance of assets by organizations is essential to realize value through managing risk and opportunity, in order to achieve the desired balance of cost, risk and performance. The regulatory and legislative environment in which organizations operate is increasingly challenging and the inherent risks that many assets present are constantly evolving.

The fundamentals of asset management and the supporting asset management system introduced in this International Standard, when integrated into the broader governance and risk framework of an organization, can contribute tangible benefits and leverage opportunities.

Asset management translates the organization's objectives into asset-related decisions, plans and activities, using a risk based approach.

2.2 Benefits of asset management

Asset management enables an organization to realize value from assets in the achievement of its organizational objectives (see [2.5.3.4](#)). What constitutes value will depend on these objectives, the nature and purpose of the organization and the needs and expectations of its stakeholders. Asset management

supports the realization of value while balancing financial, environmental and social costs, risk, quality of service and performance related to assets.

The benefits of asset management can include, but are not limited to the following:

- a) **improved financial performance:** improving the return on investments and reducing costs can be achieved, while preserving asset value and without sacrificing the short or long-term realization of organizational objectives;
- b) **informed asset investment decisions:** enabling the organization to improve its decision making and effectively balance costs, risks, opportunities and performance;
- c) **managed risk:** reducing financial losses, improving health and safety, good will and reputation, minimizing environmental and social impact, can result in reduced liabilities such as insurance premiums, fines and penalties;
- d) **improved services and outputs:** assuring the performance of assets can lead to improved services or products that consistently meet or exceed the expectations of customers and stakeholders;
- e) **demonstrated social responsibility:** improving the organization's ability to, for example, reduce emissions, conserve resources and adapt to climate change, enables it to demonstrate socially responsible and ethical business practices and stewardship;
- f) **demonstrated compliance:** transparently conforming with legal, statutory and regulatory requirements, as well as adhering to asset management standards, policies and processes, can enable demonstration of compliance;
- g) **enhanced reputation:** through improved customer satisfaction, stakeholder awareness and confidence;
- h) **improved organizational sustainability:** effectively managing short and long-term effects, expenditures and performance, can improve the sustainability of operations and the organization;
- i) **improved efficiency and effectiveness:** reviewing and improving processes, procedures and asset performance can improve efficiency and effectiveness, and the achievement of organizational objectives.

2.3 Assets

An asset is an item, thing or entity that has potential or actual value to an organization. The value will vary between different organizations and their stakeholders, and can be tangible or intangible, financial or non-financial.

The period from the creation of an asset to the end of its life is the asset life (see [3.2.2](#)). An asset's life does not necessarily coincide with the period over which any one organization holds responsibility for it; instead, an asset can provide potential or actual value to one or more organizations over its asset life, and the value of the asset to an organization can change over its asset life.

An organization may choose to manage its assets as a group, rather than individually, according to its needs, and to achieve additional benefits. Such groupings of assets may be by asset types, asset systems, or asset portfolios.

2.4 Overview of asset management

2.4.1 General

An organization's top management, employees and stakeholders should implement planning, control activities (e.g. policies, processes or monitoring actions) and monitoring activities, to exploit opportunities and to reduce risks to an acceptable level.

Asset management involves the balancing of costs, opportunities and risks against the desired performance of assets, to achieve the organizational objectives. The balancing might need to be considered over different timeframes.

Asset management enables an organization to examine the need for, and performance of, assets and asset systems at different levels. Additionally, it enables the application of analytical approaches towards managing an asset over the different stages of its life cycle (which can start with the conception of the need for the asset, through to its disposal, and includes the managing of any potential post disposal liabilities).

2.4.2 Fundamentals

Asset management is based on a set of fundamentals.

- a) **Value:** Assets exist to provide value to the organization and its stakeholders.

Asset management does not focus on the asset itself, but on the value that the asset can provide to the organization. The value (which can be tangible or intangible, financial or non-financial) will be determined by the organization and its stakeholders, in accordance with the organizational objectives.

This includes:

- 1) a clear statement of how the asset management objectives align with the organizational objectives;
- 2) the use of a life cycle management approach to realize value from assets;
- 3) the establishment of decision-making processes that reflect stakeholder need and define value.

- b) **Alignment:** Asset management translates the organizational objectives into technical and financial decisions, plans and activities.

Asset management decisions (technical, financial and operational) collectively enable the achievement of the organizational objectives.

This includes:

- 1) the implementation of risk-based, information-driven, planning and decision-making processes and activities that transform organizational objectives into asset management plans (see [2.5.3.4](#));
- 2) the integration of the asset management processes with the functional management processes of the organization, such as finance, human resources, information systems, logistics and operations;
- 3) the specification, design and implementation of a supporting asset management system.

- c) **Leadership:** Leadership and workplace culture are determinants of realization of value.

Leadership and commitment from all managerial levels is essential for successfully establishing, operating and improving asset management within the organization.

This includes:

- 1) clearly defined roles, responsibilities and authorities;
- 2) ensuring that employees are aware, competent, and empowered;
- 3) consultation with employees and stakeholders regarding asset management.

- d) **Assurance:** Asset management gives assurance that assets will fulfil their required purpose.

The need for assurance arises from the need to effectively govern an organization. Assurance applies to assets, asset management and the asset management system.

This includes:

- 1) developing and implementing processes that connect the required purposes and performance of the assets to the organizational objectives;
- 2) implementing processes for assurance of capability across all life cycle stages;
- 3) implementing processes for monitoring and continual improvement;
- 4) providing the necessary resources and competent personnel for demonstration of assurance, by undertaking asset management activities and operating the asset management system.

2.4.3 The relationship of the asset management system to asset management

An asset management system is used by the organization to direct, coordinate and control asset management activities. It can provide improved risk control and gives assurance that the asset management objectives will be achieved on a consistent basis. However, not all asset management activities can be formalized through an asset management system. For example, aspects such as leadership, culture, motivation, behaviour, which can have a significant influence on the achievement of asset management objectives, may be managed by the organization using arrangements outside the asset management system. The relationship between key asset management terms is shown in [Figure 1](#).

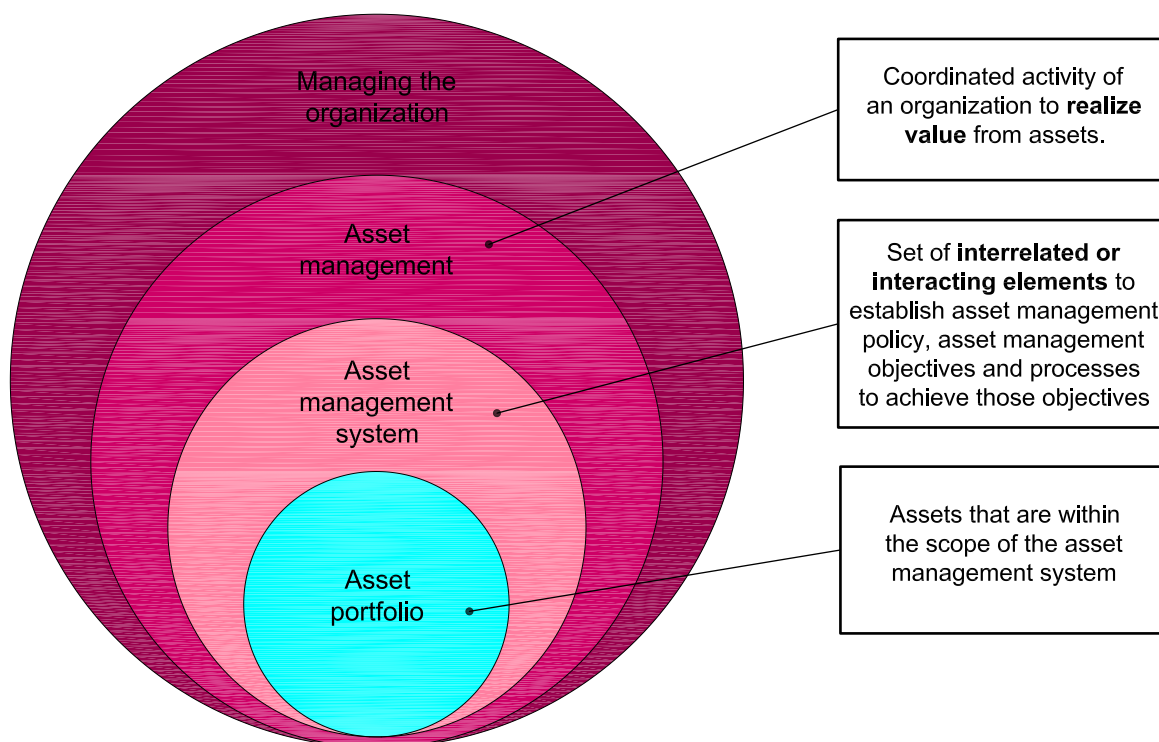


Figure 1 — Relationships between key terms

2.5 Overview of the asset management system

2.5.1 General

An asset management system is a set of interrelated and interacting elements of an organization, whose function is to establish the asset management policy and asset management objectives, and the processes, needed to achieve those objectives (see [3.4.3](#)). In this context, the elements of the asset management

system should be viewed as a set of tools, including policies, plans, business processes and information systems, which are integrated to give assurance that the asset management activities will be delivered.

Asset management requires accurate asset information, but an asset management system is more than a management information system. Asset management interacts with many functions of an organization. The assets themselves can also support more than one function and more than one functional unit within the organization. The asset management system provides a means for coordinating contributions from and interaction between these functional units within an organization.

The asset management plan can enable an organization to create a link, if needed, between its asset management system (such as described by ISO 55001, ISO 55002 and this International Standard) and a variety of specific, technical asset management requirements. These specific, technical requirements are given in standards both inside and outside the ISO environment, and at the international, regional or national standardization levels; such standards provide information on strategies and tactics, as well as specific design, construction, material or process requirements. The definitions given in 3.4 refer specifically to the asset management system described in this International Standard; some of these terms may be defined differently for a specific, technical standard.

The establishment of an asset management system is an important strategic decision for an organization. ISO 55001 specifies the requirements of an asset management system, but does not specify the design of the system. ISO 55002 provides guidance on the design and operation of an asset management system.

2.5.2 Benefits of an asset management system

An asset management system provides a structured approach for the development, coordination and control of activities undertaken on assets by the organization over different life cycle stages, and for aligning these activities with its organizational objectives.

a) Creating an asset management system provides benefits in itself.

The process of implementing an asset management system can require significant time effort and expense; however, the organization does not need to wait until the entire system is fully operational to begin accruing benefits. The benefits, or quick wins, in areas such as risk reduction, opportunity identification or process improvement can be identified early in the implementation, and can be exploited to demonstrate returns and gain stronger stakeholder support.

- Asset management is data intensive and new tools and processes are often necessary to collect, assemble, manage, analyse and use asset data. The creation and use of these tools can stimulate and improve organizational knowledge and decision making.
- The process of creating an asset management system brings new perspectives to the organization and new ideas on value creation from the use of assets. These new perspectives can also stimulate improvements in other organizational functions, such as purchasing, finance, human resources and information technology.
- The creation of an asset management system is usually cross-functional and based on life cycle considerations; this can provide a focal point for addressing the issues of functional integration of the organization and life cycle planning.

b) Top management benefits from new insights and cross functional integration.

An asset management system can help in gaining an understanding of assets, their performance, the risks associated with managing assets, investment needs, and asset value as an input to decision making and organizational strategic planning.

- Top management should recognize the need to improve communication and interaction across functions. An asset management system inherently supports this interaction. It ensures that assets are managed in an integrated manner, and asset value is improved.
- An asset management system supports a long-term and sustainable approach to decision making.

- An asset management system provides an ideal framework for the identification, understanding and integration of the many technical standards, codes, guidelines and best practices that affect the organization's assets, and support the implementation of asset management.
- An asset management system supports energy management, environmental management and other activities related to sustainability.

c) Financial functions benefit from improved data and linkages.

Integration of an organization's strategic asset management plan (SAMP) (see [2.5.3.4](#) and [3.3.2](#)), with its long-term financial plans can enable the balancing of short-term financial needs with the needs of medium-term activity plans, and with the much longer-term plans that some assets require.

- Robust financial information, based on integrated processes between the asset management and finance functions, is an important benefit of the asset management system. The linkage of asset management information to financial information is an important contribution of the asset management system to the financial function. This interaction supports improved assessment of the financial position and funding requirements of the organization in relation to its assets.
- The organization's risk-based decision making processes can become more effective by addressing asset and financial risks together, and by balancing performance, costs and risks.
- An effective taxonomy, which may be a feature of the asset management system, can enable an integrated financial and technical view of assets and asset systems.

d) Many parts of the organization benefit from an asset management system.

An asset management system touches many parts of the organization:

- the organization's human resources function may work with its asset management system on the development of competency models, training programs and processes for coaching and mentoring; these developments benefit both functions;
- some asset data comes from control systems, which are often isolated from other information systems. Integration of this data through the asset management system can provide new asset information, leading to improved organizational decision making;
- communicating with employees, suppliers and contracted service providers about the asset management system can result in improvements in the quality of asset information; it will also increase awareness amongst individuals, inside and outside of the organization, of their role in asset management decision making and the value of the activities they are undertaking;
- the asset management system can stimulate creativity and innovation by supporting people who understand the importance of asset management and are motivated to work towards achieving the asset management objectives.

2.5.3 Elements of an asset management system

2.5.3.1 General

An asset management system impacts the whole organization, including its stakeholders and external service providers, and can use, link or integrate many of the organization's activities and functions that would otherwise be managed or operated in isolation. The process of establishing an asset management system requires a thorough understanding of each of its elements and the policies, plans and procedures that integrate them.

The asset management system requirements described in ISO 55001 are grouped in a way that is consistent with the fundamentals of asset management:

- context of the organization (ISO 55001:2014, Clause 4);

- leadership (ISO 55001:2014, Clause 5);
- planning (ISO 55001:2014, Clause 6);
- support (ISO 55001:2014, Clause 7);
- operation (ISO 55001:2014, Clause 8);
- performance evaluation (ISO 55001:2014, Clause 9);
- improvement (ISO 55001:2014, Clause 10).

2.5.3.2 Context of the organization

When establishing or reviewing its asset management system, an organization should take into account its internal and external contexts. The external context includes the social, cultural, economic and physical environments, as well as regulatory, financial and other constraints. The internal context includes organizational culture and environment, as well as the mission, vision and values of the organization. Stakeholder inputs, concerns and expectations are also part of the context of the organization. The influences of stakeholders are key to setting rules for consistent decision making and also contribute to the setting of organizational objectives, which in turn, influence the design and scope of its asset management system.

2.5.3.3 Leadership

Top management is responsible for developing the asset management policy and asset management objectives and for aligning them with the organizational objectives. Leaders at all levels are involved in the planning, implementation and operation of the asset management system. Top management should create the vision and values that guide policy, practice and actively promote these values inside and outside the organization. Top management also defines the responsibilities, accountabilities and asset management objectives and strategies, which create the environment for the asset management system. Leaders should lend their authority to supporting the asset management system, and should ensure its alignment to other management systems within the organization through appropriate organizational design.

Top management and leaders at all levels are responsible for ensuring that appropriate resources are in place to support the asset management system. These resources include appropriate funding, adequate and competent human resources, and information technology support.

Leaders should recognize and resolve conflicts between the internal culture of the organization and the performance of its asset management system.

Top management and leaders at all levels are responsible for communicating the organization's asset management objectives and the importance of its asset management system to all employees, customers, suppliers, contractors and other stakeholders. Communication should be two-way, with leaders being open to receiving information aimed at improving the asset management system from all levels.

2.5.3.4 Planning

The organizational objectives provide the overarching context and direction to the organization's activities, including its asset management activities. The organizational objectives are generally produced from the organization's strategic level planning activities and are documented in an organizational plan (see [3.1.15](#)).

NOTE 1 The organizational plan can be referred to by other names, e.g. the corporate plan.

The principles by which the organization intends applying asset management to achieve its organizational objectives should be set out in an asset management policy (see [3.1.18](#)). The approach to implementing these principles should be documented in a strategic asset management plan (SAMP) (see [3.3.2](#)).

NOTE 2 A strategic asset management plan can be referred to by other names, e.g. an asset management strategy.

An organization's SAMP should be used to guide the setting of its asset management objectives, and to describe the role of the asset management system in meeting these objectives. This includes the structures, roles and responsibilities necessary to establish the asset management system and to operate it effectively. Stakeholder support, risk management and continuous improvement are important issues to be addressed in the establishment and operation of the asset management system. The SAMP can have a timeframe that extends beyond the organization's own business planning timeframe, requiring the asset management system to address the complete lifetimes of the assets.

The organization should also use its SAMP to guide its asset management system in the development of its asset management plans (i.e. in establishing what to do). The asset management plans themselves should define the activities to be undertaken on assets, and should have specific and measurable objectives (e.g. timeframes and the resources to be used). These objectives can provide the opportunity for alignment of operating plans with the organizational plan and any unit level business plans.

Aligning the asset management objectives with the organizational objectives, as well as linking asset reports to financial reports, can improve the organization's effectiveness and efficiency. The linking of asset reports to financial reports can also improve and clarify the assessment of the financial status and long-term funding requirements of the organization.

2.5.3.5 Support

The asset management system will require collaboration among many parts of the organization. This collaboration often involves the sharing of resources. Coordinating these resources and applying, verifying and improving their use should be objectives of the asset management system. It should also promote awareness of the asset management objectives across the whole organization.

The asset management system provides information to support the development of asset management plans and the evaluation of their effectiveness. Asset information systems can be extremely large and complex in some organizations, and there are many issues involved in collecting, verifying and consolidating asset data in order to transform it into asset information. Creating, controlling, and documenting this information is a critical function of the asset management system.

The asset management system should specify the competency requirements for personnel involved in asset management. The implementation, maintenance, evaluation and improvement of these competencies normally requires close cooperation with the organization's human resource management system. These two systems should be mutually supportive.

2.5.3.6 Operation

The organization's asset management system can enable the directing, implementation and control of its asset management activities, including those that have been outsourced. Functional policies, technical standards, plans and processes for the implementation of the asset management plans should be fed back into the design and operation of the asset management system.

Operation of the asset management system can sometimes require planned changes to asset management processes or procedures, which can introduce new risks. Risk assessment and control in the context of managing change is an important consideration in operating an asset management system.

When an organization outsources some of its asset management activities, this should not remove those outsourced activities from the control of the organization's asset management system. In situations where interacting activities are outsourced to different service providers, the responsibilities and complexity of control will be increased.

2.5.3.7 Performance evaluation

The organization should evaluate the performance of its assets, its asset management and its asset management system. Performance measures can be direct or indirect, financial or non-financial.

Asset performance evaluation is often indirect and complex. Effective asset data management and the transformation of data to information (see 2.5.3.5) is a key to measuring asset performance. Monitoring, analysis and evaluation of this information should be a continuous process. Asset performance evaluations should be conducted on assets managed directly by the organization and on assets which are outsourced.

Asset management performance should be evaluated against whether the asset management objectives have been achieved, and if not, why not. Where applicable, any opportunities that arose from having exceeded the asset management objectives should also be examined, as well as any failure to realize them. The adequacy of the decision-making processes should be examined carefully.

The performance of the asset management system should be evaluated against any objectives set specifically for the system itself (either when it was established, or following previous evaluations). The primary purpose of evaluating the system should be to determine whether it is effective and efficient in supporting the organization's asset management. Periodic audits should be used to evaluate the performance of the asset management system; these may be complemented by self-assessments.

The results of performance evaluations should be used as inputs into management reviews.

2.5.3.8 Improvement

An organization's asset management system is likely to be complex and continually evolving to match its context, organizational objectives and its changing asset portfolio. Continual improvement is a concept that is applicable to the assets, the asset management activities and the asset management system, including those activities or processes which are outsourced.

Opportunities for improvement can be determined directly through monitoring the performance of the asset management system, and through monitoring asset performance.

Nonconformities or potential nonconformities of the asset management system can also be identified through management reviews and internal or external audits. The nonconformities require corrective action and the potential nonconformities require preventive action.

Of particular importance are asset-related incidents or emergency situations, for which emergency response planning and business continuity planning for identified risks should be addressed by the asset management system. All such incidents, including unanticipated events, should be investigated and reviewed to see if any improvements are needed to the asset management system, to prevent their recurrence and to mitigate their effects.

Improvements should be risk assessed prior to being implemented.

2.6 Integrated management systems approach

Using an integrated management systems approach allows an organization's asset management system to be built on elements of its other management systems, such as for quality, environment, health and safety, and risk management. Building on existing systems can reduce the effort and expense involved in creating and maintaining an asset management system. It can also improve integration across different disciplines and improve cross-functional coordination.

Organizations that have implemented an integrated systems approach have demonstrated the benefits of the integrated approach and shortened the time to implementation of each new system. The integrated approach, in addition to reducing cost, reduces risks and improves acceptance of each new system.

Asset management, because it touches so many parts of the organization, is a natural candidate for an integrated systems approach.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1 General terms

3.1.1

audit

systematic, independent and documented *process* (3.1.19) for obtaining audit evidence and evaluating it objectively to determine the extent to which the audit criteria are fulfilled

Note 1 to entry: An audit can be an internal audit (first party) or an external audit (second party or third party), and it can be a combined or integrated audit (combining two or more disciplines).

Note 2 to entry: "Audit evidence" and "audit criteria" are defined in ISO 19011.

3.1.2

capability

<**asset management**> measure of capacity and the ability of an entity (system, person or *organization* (3.1.13)) to achieve its *objectives* (3.1.12)

Note 1 to entry: *Asset management* (3.3.1) capabilities include *processes* (3.1.19), resources, *competences* (3.1.3) and technologies to enable the effective and efficient development and delivery of *asset management plans* (3.3.2) and *asset life* (3.2.2) activities, and their *continual improvement* (3.1.5).

3.1.3

competence

ability to apply knowledge and skills to achieve intended results

3.1.4

conformity

fulfilment of a *requirement* (3.1.20)

3.1.5

continual improvement

recurring activity to enhance *performance* (3.1.17)

3.1.6

documented information

information required to be controlled and maintained by an *organization* (3.1.13) and the medium on which it is contained

Note 1 to entry: Documented information can be in any format and media and from any source.

Note 2 to entry: Documented information can refer to:

- the *management system* (3.4.2), including related *processes* (3.1.19);
- information created in order for the organization to operate (documentation);
- evidence of results achieved (e.g. records, key performance indicators).

3.1.7

effectiveness

extent to which planned activities are realized and planned results achieved

3.1.8

incident

unplanned event or occurrence resulting in damage or other loss

3.1.9**monitoring**

determining the status of a system, a *process* (3.1.19) or an activity

Note 1 to entry: To determine the status, there may be a need to check, supervise or critically observe.

Note 2 to entry: For the purposes of asset management, monitoring may also refer to determining the status of an asset. This is typically referred to as “condition monitoring” or “performance monitoring”.

3.1.10**measurement**

process (3.1.19) to determine a value

3.1.11**nonconformity**

non-fulfilment of a *requirement* (3.1.20)

Note 1 to entry: Nonconformity can be any deviation from asset *management system* (3.4.2) requirements, or from relevant work standards, practices, procedures, legal requirements, etc.

3.1.12**objective**

result to be achieved

Note 1 to entry: An objective can be strategic, tactical or operational.

Note 2 to entry: Objectives can relate to different disciplines (such as financial, health and safety, and environmental goals) and can apply at different levels (such as strategic, organization-wide, project, product and *process* (3.1.19)).

Note 3 to entry: An objective can be expressed in other ways, e.g. as an intended outcome, a purpose, an operational criterion, an *asset management* (3.3.1) objective or by the use of other words with similar meaning (e.g. aim, goal, or target).

Note 4 to entry: In the context of *asset management systems* (3.4.3), asset management objectives are set by the *organization* (3.1.13), consistent with the *organizational objectives* (3.1.14) and *asset management policy* (3.1.18), to achieve specific measurable results.

3.1.13**organization**

person or group of people that has its own functions with responsibilities, authorities and relationships to achieve its *objectives* (3.1.12)

Note 1 to entry: The concept of organization includes, but is not limited to, sole-trader, company, corporation, firm, enterprise, authority, partnership, charity or institution, or part or combination thereof, whether incorporated or not, public or private.

3.1.14**organizational objective**

overarching *objective* (3.1.12) that sets the context and direction for an *organization's* (3.1.13) activities

Note 1 to entry: Organizational objectives are established through the strategic level planning activities of the organization.

3.1.15**organizational plan**

documented information (3.1.6) that specifies the programmes to achieve the *organizational objectives* (3.1.14)

3.1.16

outsource (verb)

make an arrangement where an external *organization* (3.1.13) performs part of an organization's function or *process* (3.1.19)

Note 1 to entry: An external organization is outside the scope of the *management system* (3.4.2), although the outsourced function or process is within the scope if its activities influence the effectiveness of the *asset management system* (3.4.3).

3.1.17

performance

measurable result

Note 1 to entry: Performance can relate either to quantitative or qualitative findings.

Note 2 to entry: Performance can relate to the management of activities, *processes* (3.1.19), products (including services), systems or *organizations* (3.1.13).

Note 3 to entry: For the purposes of *asset management* (3.3.1), performance can relate to *assets* (3.2.1) in their ability to fulfil *requirements* (3.1.20) or *objectives* (3.1.12).

3.1.18

policy

intentions and direction of an *organization* (3.1.13) as formally expressed by its *top management* (3.1.23)

3.1.19

process

set of interrelated or interacting activities which transforms inputs into outputs

3.1.20

requirement

need or expectation that is stated, generally implied or obligatory

Note 1 to entry: "Generally implied" means that it is custom or common practice for the *organization* (3.1.13) and *stakeholders* (3.1.22) that the need or expectation under consideration is implied.

Note 2 to entry: A specified requirement is one that is stated, for example in *documented information* (3.1.6).

3.1.21

risk

effect of uncertainty on *objectives* (3.1.12)

Note 1 to entry: An effect is a deviation from the expected — positive and/or negative.

Note 2 to entry: Objectives can relate to different disciplines (such as financial, health and safety, and environmental goals) and can apply at different levels (such as strategic, organization-wide, project, product and *process* (3.1.19)).

Note 3 to entry: Risk is often characterized by reference to potential "events" (as defined in ISO Guide 73:2009, 3.5.1.3) and "consequences" (as defined in ISO Guide 73:2009, 3.6.1.3), or a combination of these.

Note 4 to entry: Risk is often expressed in terms of a combination of the consequences of an event (including changes in circumstances) and the associated "likelihood" (ISO Guide 73:2009, 3.6.1.1) of occurrence.

Note 5 to entry: Uncertainty is the state, even partial, of deficiency of information related to, understanding or knowledge of, an event, its consequence, or likelihood.

[SOURCE: ISO Guide 73:2009, 1.1]

3.1.22

stakeholder

person or *organization* (3.1.13) that can affect, be affected by, or perceive themselves to be affected by a decision or activity

Note 1 to entry: A "stakeholder" can also be referred to as an "interested party".

3.1.23**top management**

person or group of people who directs and controls an *organization* (3.1.13) at the highest level

Note 1 to entry: Top management has the power to delegate authority and provide resources within the organization.

Note 2 to entry: If the scope of the *management system* (3.4.2) covers only part of an organization, then top management refers to those who direct and control that part of the organization. If multiple *asset management systems* (3.4.3) are employed, the systems should be designed to coordinate efforts.

3.2 Terms relating to assets**3.2.1****asset**

item, thing or entity that has potential or actual value to an *organization* (3.1.13)

Note 1 to entry: Value can be tangible or intangible, financial or non-financial, and includes consideration of *risks* (3.1.21) and liabilities. It can be positive or negative at different stages of the *asset life* (3.2.2).

Note 2 to entry: Physical assets usually refer to equipment, inventory and properties owned by the organization. Physical assets are the opposite of intangible assets, which are non-physical assets such as leases, brands, digital assets, use rights, licences, intellectual property rights, reputation or agreements.

Note 3 to entry: A grouping of assets referred to as an *asset system* (3.2.5) could also be considered as an asset.

3.2.2**asset life**

period from *asset* (3.2.1) creation to asset end-of-life

3.2.3**life cycle**

stages involved in the management of an *asset* (3.2.1)

Note 1 to entry: The naming and number of the stages and the activities under each stage usually vary in different industry sectors and are determined by the *organization* (3.1.13).

3.2.4**asset portfolio**

assets (3.2.1) that are within the scope of the *asset management system* (3.4.3)

Note 1 to entry: A portfolio is typically established and assigned for managerial control purposes. Portfolios for physical hardware might be defined by category (e.g. plant, equipment, tools, land). Software portfolios might be defined by software publisher, or by platform (e.g. PC, server, mainframe).

Note 2 to entry: An asset management system can encompass multiple asset portfolios. Where multiple asset portfolios and asset management systems are employed, *asset management* (3.3.1) activities should be coordinated between the portfolios and systems.

3.2.5**asset system**

set of *assets* (3.2.1) that interact or are interrelated

3.2.6**asset type**

grouping of *assets* (3.2.1) having common characteristics that distinguish those assets as a group or class

EXAMPLE Physical assets, information assets, intangible assets, *critical assets* (3.2.7), enabling assets, linear assets, information and communications technology (ICT) assets, infrastructure assets, moveable assets.

3.2.7

critical asset

asset (3.2.1) having potential to significantly impact on the achievement of the *organization's* (3.1.13) *objectives* (3.1.12)

Note 1 to entry: Assets can be safety-critical, environment-critical or *performance-critical* (3.1.17) and can relate to legal, regulatory or statutory *requirements* (3.1.20).

Note 2 to entry: Critical assets can refer to those assets necessary to provide services to critical customers.

Note 3 to entry: *Asset systems* (3.2.5) can be distinguished as being critical in a similar manner to individual assets.

3.3 Terms relating to asset management

3.3.1

asset management

coordinated activity of an *organization* (3.1.13) to realize value from *assets* (3.2.1)

Note 1 to entry: Realization of value will normally involve a balancing of costs, *risks* (3.1.21), opportunities and *performance* (3.1.17) benefits.

Note 2 to entry: Activity can also refer to the application of the elements of the *asset management system* (3.4.3).

Note 3 to entry: The term “activity” has a broad meaning and can include, for example, the approach, the planning, the plans and their implementation.

3.3.2

strategic asset management plan

SAMP

documented information (3.1.6) that specifies how *organizational objectives* (3.1.14) are to be converted into *asset management* (3.3.1) *objectives* (3.1.12), the approach for developing *asset management plans* (3.3.3), and the role of the *asset management system* (3.4.3) in supporting achievement of the asset management objectives

Note 1 to entry: A strategic asset management plan is derived from the *organizational plan* (3.1.15).

Note 2 to entry: A strategic asset management plan may be contained in, or may be a subsidiary plan of, the organizational plan.

3.3.3

asset management plan

documented information (3.1.6) that specifies the activities, resources and timescales required for an individual *asset* (3.2.1), or a grouping of assets, to achieve the *organization's* (3.1.13) *asset management* (3.3.1) *objectives* (3.1.12)

Note 1 to entry: The grouping of assets may be by *asset type* (3.2.6), asset class, *asset system* (3.2.5) or *asset portfolio* (3.2.4).

Note 2 to entry: An asset management plan is derived from the *strategic asset management plan* (3.3.2).

Note 3 to entry: An asset management plan may be contained in, or may be a subsidiary plan of, the strategic asset management plan.

3.3.4

preventive action

action to eliminate the cause of a potential *nonconformity* (3.1.11) or other undesirable potential situation

Note 1 to entry: This definition is specific to *asset management* (3.3.1) activities only.

Note 2 to entry: There can be more than one cause for a potential nonconformity.

Note 3 to entry: Preventive action is taken to prevent occurrence and to preserve an *asset's* (3.2.1) function, whereas *corrective action* (3.4.1) is taken to prevent recurrence.

Note 4 to entry: Preventive action is normally carried out while the asset is functionally available and operable or prior to the initiation of functional failure.

Note 5 to entry: Preventive action includes the replenishment of consumables where the consumption is a functional *requirement* (3.1.19).

[SOURCE: ISO 9000:2005, 3.6.4, modified – Note 3 to entry has been modified; Notes 1, 4 and 5 have been added]

3.3.5

predictive action

action to monitor the condition of an *asset* (3.2.1) and predict the need for *preventive action* (3.3.4) or *corrective action* (3.4.1)

Note 1 to entry: Predictive action is also commonly referred to as either “condition monitoring” or “performance monitoring”.

3.3.6

level of service

parameters, or combination of parameters, which reflect social, political, environmental and economic outcomes that the *organization* (3.1.13) delivers

Note 1 to entry: The parameters can include safety, customer satisfaction, quality, quantity, capacity, reliability, responsiveness, environmental acceptability, cost and availability.

3.4 Terms relating to asset management system

3.4.1

corrective action

action to eliminate the cause of a *nonconformity* (3.1.11) and to prevent recurrence

Note 1 to entry: In the case of other undesirable outcomes, action is necessary to minimize or eliminate the causes and to reduce the impact or prevent recurrence. Such actions fall outside the concept of corrective action, in the sense of this definition.

3.4.2

management system

set of interrelated or interacting elements of an *organization* (3.1.13) to establish *policies* (3.1.18) and *objectives* (3.1.12) and *processes* (3.1.19) to achieve those objectives

Note 1 to entry: A management system can address a single discipline or several disciplines.

Note 2 to entry: The system elements include the organization's structure, roles and responsibilities, planning, operation, etc.

Note 3 to entry: The scope of a management system may include the whole of the organization, specific and identified functions of the organization, specific and identified sections of the organization, or one or more functions across a group of organizations.

3.4.3

asset management system

management system (3.4.2) for *asset management* (3.3.1) whose function is to establish the asset management *policy* (3.1.18) and asset management *objectives* (3.1.12)

Note 1 to entry: The asset management system is a subset of asset management.

Annex A **(informative)**

Information on asset management activities

Relevant asset management subject areas addressed by other published international, regional, or national standards include, but are not limited to, the following:

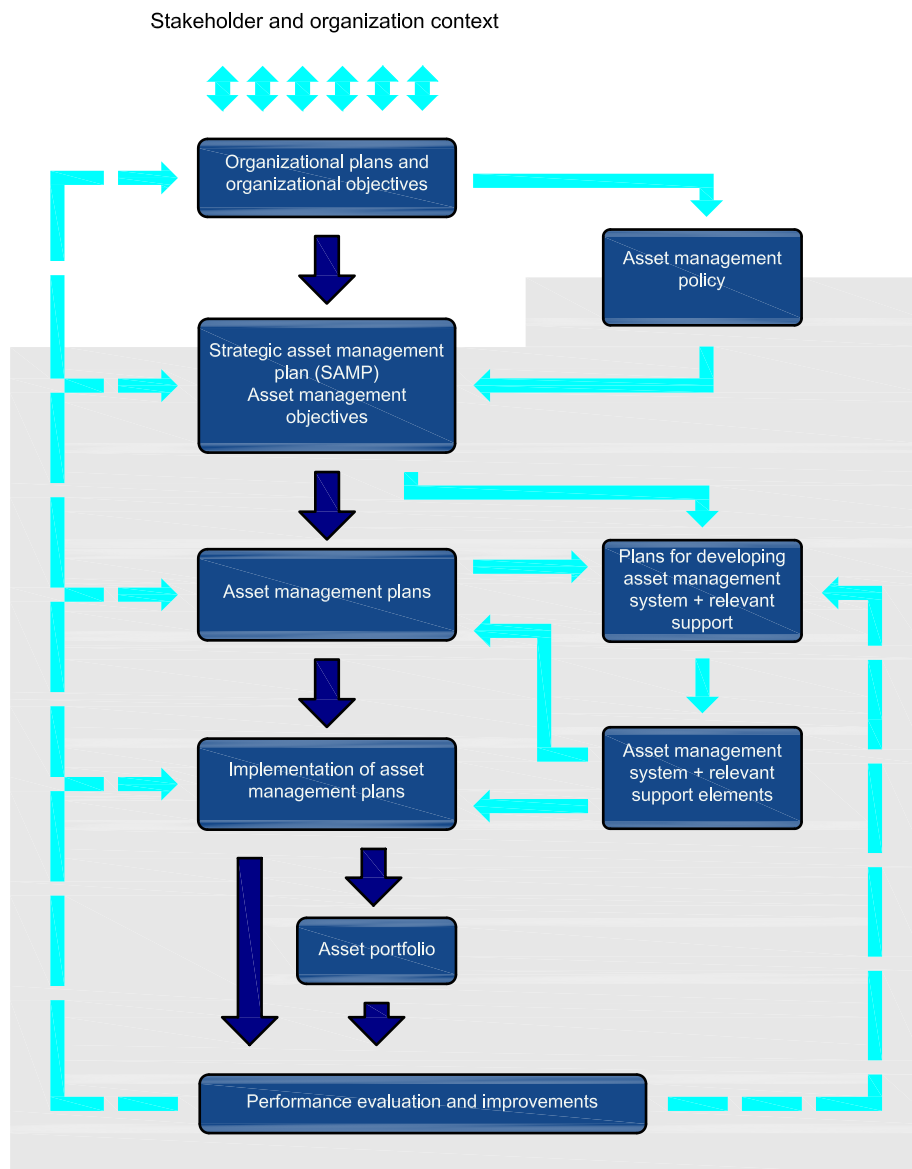
- data management;
- condition monitoring;
- risk management;
- quality management;
- environmental management;
- systems and software engineering;
- life cycle costing;
- dependability (availability, reliability, maintainability, maintenance support);
- configuration management;
- tero-technology;
- sustainable development;
- inspection;
- non-destructive testing;
- pressure equipment;
- financial management;
- value management;
- shock and vibration;
- acoustics;
- qualification and assessment of personnel;
- project management;
- property and property management;
- facilities management;
- equipment management;
- commissioning process;
- energy management.

Users of ISO 55001, ISO 55002 and this International Standard should also refer to such standards wherever possible, to ensure consistent delivery of asset management throughout their organization.

Annex B (informative)

Relationship between key elements of an asset management system

Figure B.1 shows the relationship between key elements of an asset management system.



NOTE The grey highlighted box designates the boundary of the asset management system.

Figure B.1 — Relationship between key elements of an asset management system

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